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FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

EDGE COMPUTING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define edge computing	L1	CO1	**02**
	b.	Mention the MQQT data types	L1	CO3	**02**
	c.	State Metcalfe and Beckstrome's law With an equation	L5	CO4	**02**
	d.	What are the characteristics of MQTT	L4	CO3	**02**
	e.	What are Applications of Fog Computing	L4	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the Role of Cloud Computing on the Internet of Things	L1	CO2	**05**
		OR			
	b.	With a need diagram explain the working of fog computing	L1	CO2	**05**
3.	**a.**	List out the difference between IoT and M2M with example	L2	CO2	**05**
		OR			
	b.	Write a brief note on IoT edge computing architecture	L2	CO2	**05**
4.	**a.**	What is Cloud Computing? Explain how raspberry pi considered as edge computing device with an use case	L4	CO3	**05**
		OR			
	b.	With a neat diagram explain the edge computing project using raspberry pi.	L4	CO3	**05**
5.	**a.**	How does MQTT work? Explain with an use case.	L4	CO4	**05**
		OR			
	b.	What is the difference between HTTP and MQTT?	L5	CO4	**05**
6.	**a.**	Explain the steps to set up Raspberry Pi to develop an Edge computing project	L5	CO3	**05**
		OR			
	b.	With an use case explain the concept of Industrial IoT using Raspberry pi	L5	CO5	**05**

7.	**a.**	Discuss how to access a Raspberry Pi through an SSH server	L4	CO3	**05**
		OR			
	b.	Discuss the Applications of MQTT	L4	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	With an use case explain the concept of Internet of Things	L4	CO2	**10**
		OR			
	b.	With a neat diagram explain the concept of edge computing	L5	CO2	**10**
9.	**a.**	Write a program to interface DHT sensor with and Raspberry Pi	L5	CO3	**10**
		OR			
	b.	Explain the concept of IoT and M2M and list out the differences between IOT and M2M	L4	CO2	**10**
10.	**a.**	Explain 10 working features of Raspberry pi	L2	CO3	**10**
		OR			
	b.	Explain the concept of edge computing using raspberry pi with an use case	L2	CO3	**10**
11.	**a.**	Explain the MQTT architecture	L4	CO4	**10**
		OR			
	b.	Discuss the application and characteristics of MQTT	L4	CO4	**10**
12.	**a.**	With an use case explain how the edge devices are use full in health care sector	L5	CO5	**10**
		OR			
	b.	Explain how edge devices are connected in Industrial IoT and Commercial IoT with examples	L5	CO5	**10**
13.	**a.**	With an use case explain the MQTT Packet format	L4	CO4	**10**
		OR			
	b.	Explain the architecture of cloud computing	L4	CO2	**10**